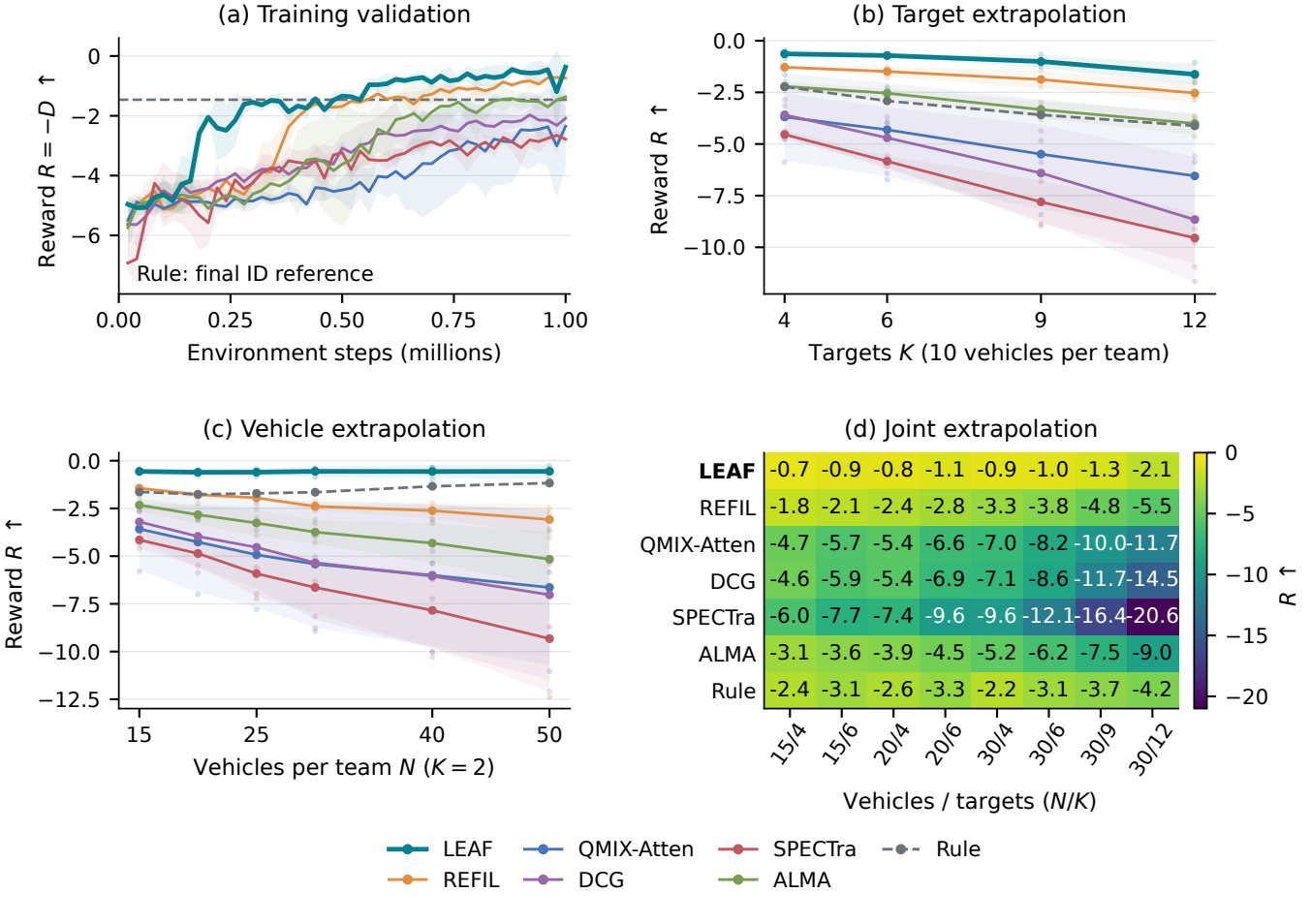

(a) Training validation
Reward $R = -D$ ↑
0
−2
−4
−6
Rule: final ID reference
0.00
0.25
0.50
0.75
1.00
Environment steps (millions)
(b) Target extrapolation
Reward R ↑
0.0
−2.5
−5.0
−7.5
−10.0
4
6
9
12
Targets K (10 vehicles per team)
(c) Vehicle extrapolation
Reward R ↑
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
15
25
40
50
Vehicles per team N ($K = 2$)
(d) Joint extrapolation
LEAF -0.7 -0.9 -0.8 -1.1 -0.9 -1.0 -1.3 -2.1
REFIL -1.8 -2.1 -2.4 -2.8 -3.3 -3.8 -4.8 -5.5
QMIX-Atten -4.7 -5.7 -5.4 -6.6 -7.0 -8.2 -10.0 -11.7
DCG -4.6 -5.9 -5.4 -6.9 -7.1 -8.6 -11.7 -14.5
SPECTra -6.0 -7.7 -7.4 -9.6 -9.6 -12.1 -16.4 -20.6
ALMA -3.1 -3.6 -3.9 -4.5 -5.2 -6.2 -7.5 -9.0
Rule -2.4 -3.1 -2.6 -3.3 -2.2 -3.1 -3.7 -4.2
15/4
15/6
20/4
20/6
30/4
30/6
30/9
30/12
Vehicles / targets (N/K)
0
−5
−10
−15
−20
R ↑
LEAF
REFIL
QMIX-Atten
DCG
SPECTra
ALMA
Rule